Xinxin Wang

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Education

- **Case Western Reserve University (CWRU) and Cleveland Clinic (CC)** Aug 2023 - Now
  PhD candidate of Computer Science; **GPA:** 4.0/4.0; **Advisors:** Prof. Xiaojuan Li and Prof. Jing Li
- **Sun Yat-sen University (SYSU)** Aug 2019 - June 2023
  Bachelor of Engineering in Intelligence Science and Technology; **GPA:** 3.7/4.0; **Advisor:** Prof. Shen Zhao

Internship

- **Index Engines** May 2025 - Aug 2025
  Machine Learning Engineer; under the mentorship of *Jairo Esteban*;
  - **Ransomware detection model**: Designed and implemented an end-to-end machine learning pipeline for ransomware detection. Responsibilities included data curation, literature review of recent academic research, experimentation with various model architectures, and optimization of hyperparameters for improved performance.
  - **Agile development**: Collaborated in a cross-functional team using Agile methodology. Participated in sprint planning, daily stand-ups, and regular retrospectives. Contributed to iterative development, task tracking, and milestone-based project delivery.
- **Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** Apr 2023 - Jul 2023
  Research Assistant at Medical Robotics Center; under the guidance of *Dr. Zhixiong Yang*;
  - **Improve segmentation workflow**: Establish the pipeline of pulmonary airway and artery-vein segmentation from CT scans, for visualization and navigation of our self-developed lung intervention surgery platform.
- **Hao Yi Environmental Protection and Energy Storage Technology Co., Ltd.** Feb 2023 - Mar 2023
  Assistant Product Manager of Battery Passport Development;
  - **Launching Ebatte Passport**: Read the latest EU Battery Regulation, outline the full lifecycle of the digital battery passport, and further extract the requirements for manufacturers exporting to the EU that must be met regarding the battery passport to assist in advancing the Ebatte product.

Academic Research

- **Foundation model for 3D Knee MRI Analysis** Mar 2024 - Now
  First author; collaborating with IBM team; under the guidance of *Prof. Xiaojuan Li and Prof. Mingrui Yang*;
  - **Manuscript**: "KneePreM:A Self-Supervised Pretrained Model for Robust and Efficient 3D Knee MRI Analysis" for submission to *Radiology AI*
  - **Publication**: Abstract accepted to the conference of International Society for Magnetic Resonance in Medicine (ISMRM)
  - **Responsibility**: Help IBM team with the understanding of Osteoarthrits Initiative database (OAI); conduct experiments and validate code deliveries from IBM side; provide clinical insights and contribute to the OAI example of an open source code platform FUSEMEDML(link); transfer to in-house data validation on HPC server.
  - **Future work**: Develop musculoskeletal foundation model with more clinical cohorts and ViT architecture
- **Reliable Lesion Segmentation with Calibrated Uncertainty** Aug 2023 - Mar 2025
  First author; under the guidance of *Prof. Xiaojuan Li* and *Prof. Shuo Li*;
  - **Publication**: "CalDiff: calibrating uncertainty and accessing reliability of diffusion models for trustworthy lesion segmentation" is accepted by IEEE Journal of Biomedical and Health Informatics (IEEE-JBHI).
  - **Innovation**: Caldiff is proposed with uncertainty calibrated on both step-wise and sequence-aware level, providing calibrated uncertainty maps generated from multiple plausible segmentation results, which can serve as a robust tool for lesion segmentation in clinical practice.
- **Effcient Uncertainty-aware Lesion Segmentation** Oct 2021 - June 2023
  Joint First author; under the guidance of *Prof. Shen Zhao*;
  - **Publication**: "Customized T-time Inner Sampling Network with Uncertainty-aware Data Augmentation Strategy for Multi-annotated Lesion Segmentation" accepted by *Computers in Biology and Medicine* (link).
  - **Innovation**: An efficient and flexible probabilistic model architecture with a low-cost data augmentation strategy devised from multi-annotated datasets to adaptively capture both high and low degree of uncertainty in lesion segmentations.
- **Deep Active Contour Model for Vertebrae Segmentation** Nov 2022 - June 2023
  Third author; under the guidance of *Prof. Shen Zhao*;
  - **Publication**: "Attractive Deep Morphology-aware Active Contour Network for Vertebral Body Contour Extraction in Heterogeneous MR " accepted by *Medical Image Analysis (MedIA)* (link).

- **Contribution**: Investigating latest literature on semantic segmentation methods for comparison with proposed contour-based segmentation model; preprocessing publicly accessible datasets; carrying out additional comparison experiments.

- **Boundary-aware Osteosarcoma Segmentation based on Multi-modality MRIs** Feb 2023 - May 2023
  Third Author; under the guidance of *Prof. Shen Zhao*;
  - **Publication**: "DECIDE: A decoupled semantic and boundary learning network for precise osteosarcoma segmentation by integrating multi-modality MRI" accepted by *Computers in Biology and Medicine* (link).
  - **Innovation**: Context Attention Module is introduced into the existing segmentation framework to form a prediction module of tumor contour, which was supervised and learned in training to refine segmentation results.

Project Design

- **Dorm Life Helper - a WeChat Small Program** Spring semester of 2022
  Group leader; coursework of Software Engineering;
  - **App Launched**: *Dorm Life Helper* was developed in JavaScript and launched on WeChat small program platform. It can be searched in WeChat and is available to provide convenient services related to dorm life.
  - **Modules**: Login Module - login, create/join/exit from/dissolve a dorm; Member Module - check/edit dorm's info, check/edit members' info; Maintaining Module - duty rotation, bill record, cost splitting; Moments Module: add/edit/delete shared memories; Community Module - post advertisements/messages/comments.

- **Retracing the Long March - a 3D Shooter Game** Fall semester of 2021
  Group leader; coursework of VR and AR Technology;
  - **Software Copyright Registered**: The software copyright registration was approved by Copyright Protection Center of China. (Registration Number: 2022SR0438810)
  - **Innovation**: A creative 3D game developed with C# in Unity, combining quiz with shooting, set in three historic scenes.

- **GNN-based Diagnostic Reports System for Skin Diseases** July 2021 - Sep 2021
  Group leader; under the guidance of *Prof. Shancheng Jiang*;
  - **Prize Won**: Silver Award of National College Students' 'Internet+' Innovation and Entrepreneurship Competition in Guangdong Province.
  - **Innovation**: An applicable diagnostic reports generation system for dermoscopy image data, including an edge-labeling graph neural network tackling the problem of few-shot learning, and a basic LSTM language model for text generation.

- **Selfie2anime - a Real-time Style Transfer Demo** Spring semester of 2021
  Group member; coursework of Deep Learning;
  - **Innovation**: A real-time demo developed in JupyterLab; Face Alignment employed to refine the result generated from Unsupervised Generative Attentional Networks with Adaptive Layer-Instance Normalization.

Teaching Assistantship

- **CSDS413: Introduction to Data Analysis, CWRU** Jan 2026 - Now
  Teaching Assistant; Instructor: *Prof. Jing Li*;

- **ISE357: Comprehensive Experiments of Intelligent Robot, SYSU** Sep 2022 - Oct 2022
  Teaching Assistant; Instructor: *Prof. Jianqing Peng*;

Community Activity

- **A Speech Contest Hosted for All Freshmen of SYSU** Aug 2021 - Sep 2021
  Chief planner; the first speech contest at SYSU directly facing nearly 8000 newcomers;

- **Human Cyber Physical Intelligence Integration Club** Oct 2021 - Sep 2022
  Student Member; founded by School of Computer Science and Engineering, SYSU;

- **AI Boot Camp for Middle School Students (Online)** Jan 2021 - Feb 2021
  Volunteer instructor; organized by School of Intelligent Systems Engineering, SYSU;

- **Enrollment Promotion of SYSU** Jan 2020 - Jan 2020
  Excellent Volunteer awarded;

Skills Summary

- **Programming Languages**: Python, Matlab, Javascript, Latex, C#, C, C++
- **Tools**: Slurm, Git, Origin, Pycharm, Vscode, VMware, Unity, AutoCAD, Creo
- **Software Engineering**: Agile/Scrum, End-to-end ML Pipelines, Full-stack WeChat Mini-programs, API Development, UI/UX Prototyping
- **Medical AI Specialty**: 3D Medical Imaging (MRI/CT), Lesion Segmentation, Radiomics, Clinical Data Validation, Medical Robots